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Unit - 2

Fuel

Q1 Define Fuels and classify the coal on the basis of carbon content.

Ans Any material which produces a large amount of heat energy on complete combustion, the heat energy can be utilized for domestic & industrial purpose. This material is known as Fuel.

On the basis of carbon content, coal has four types -

(i) Peat => 57 to 60% of carbon presence lowest category of coal

(ii) Lignite => 60 to 70% of carbon presence more compact than peat.

(iii) Bituminous => 75 to 90% of carbon presence most commonly used coal.

(iv) Anthracite => 90 to 95% of carbon presence best quality of coal.

Q2 What are good fuel's characteristics?

Ans A Good fuel contains -

(i) low quantity of moisture, volatile matter.

(ii) high calorific value, burns easily without producing smoke or harmful gases.

(iii) Should be sustainable & renewable.

(iv) Should be safe & pollution free

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Q3 What are gross & net calorific values?

Ans Net calorific value is the amount of heat produced after combustion, when volatile matter / gases are allowed to escape in the atmosphere (LCV)

Gross calorific value is the amount of heat produced by combustion of fuel & gases & volatile matter is allowed to coal is known as gross calorific value. (HCV)

$$\boxed{HCV = LCV + \text{Heat of vaporization}}$$

Q4 What is pulverisation of coal? write its significance

Ans Coal is converted into powder form is known as pulverisation.

It increase surface area of coal & burning becomes easier.

Q5 Explain ultimate ~~process~~ Analysis of coal.

Ans It consists of determination of combustion value ash, moisture, volatile metter.

* Moisture Content:- Powdered coal is taken in silica crucible & heated & cooling for 1Hr.

$$\% \text{ of moisture} = \frac{\text{loss in weight}}{\text{weight of coal}} \times 100$$

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(ii) Volatile content :- moisture free coal heated in muffle furnace for 7 to 8 minute.

$$\% \text{ volatile} = \frac{\text{loss in weight}}{\text{weight of coal}} \times 100$$

(iii) Ash :- Volatile free coal heated at 700-750°C in presence of oxygen

$$\% \text{ ash} = \frac{\text{weight of ash}}{\text{weight of coal}} \times 100$$

(iv) Fixed Carbon :- ~~Volatile free coal heated~~. Remaining material, after removel of volalite metter, Ash, moisture, is known as fixed carbon.

$$\% \text{ Fixed carbon} = 100 - (\% \text{ moisture} + \% \text{ volatile} + \% \text{ Ash})$$

Significance :-

It provide information about quality of coal.

→ If % of fixed carbon is high then it has high calorific value which mean it's a good quality coal.

→ If % of moisture & volatile metter is low then It's a good quality of coal.

→ If Ash is less the coal is of a good quality

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Q6 What is carbonation? Explain recovery of by product of Otto - Hafmann's by product method.

Ans Coal is heated strongly in absence of air, it converts into coke.

Coke is better then coal.

$$\text{Coal} \xrightarrow[\text{absence of air}]{\text{heat}} \text{Coke}$$

It is known as carbonation.

* Recovery of by product of otto - Haffmann's by pro method

It contains tar, amonia, H_2S, naphalene and moisture.

1. Tar :- Coke ~~process~~ passed through NH_3 (Ammonia gas) and it saperate tar.

(2) Ammonia :- by sprading water.

$$NH_3 + H_2O \longrightarrow NH_4OH$$

Now

$$NH_4OH + H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

↓

used as fertilizer.

(3) Napthalene :- The gases are passed where water at low temperature is sprayed.

Napthalene get condensed.

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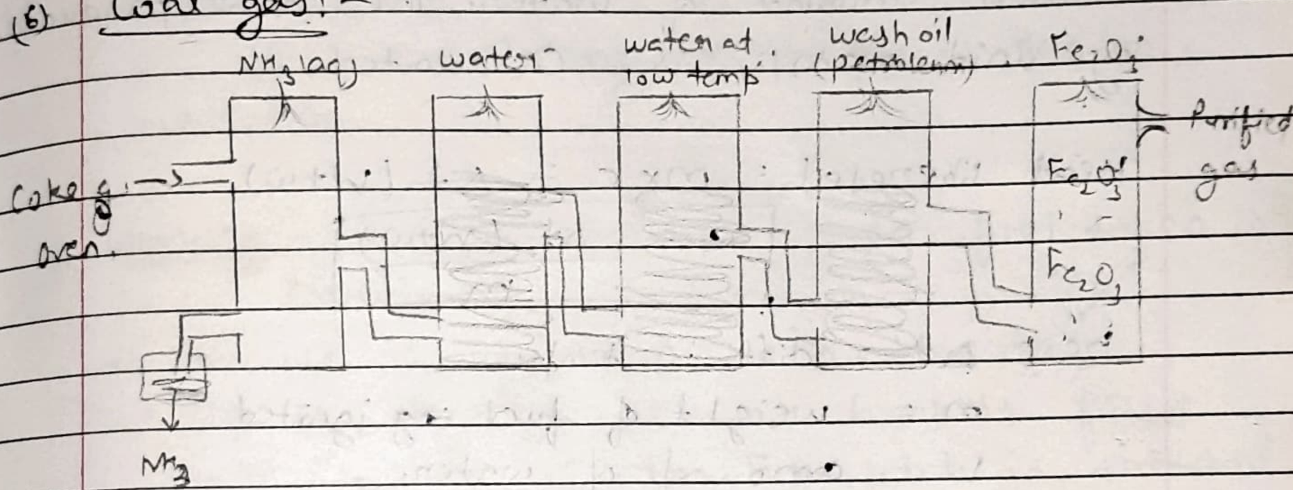
(4) Benzene :- The rest part is passed through petroleum.

(5) H_2S :- Passing through moist Fe_2O_3

$$Fe_2O_3 + 3H_2O \longrightarrow Fe_2S_3 + 3H_2O$$

(6) Coal gas :-



Q7 Why is coke preferred in metallurgical process?

Ans Coke is preferred in metallurgical process because

(i) It has high carbon content. It helps in reducing metal oxides to their respective values

(ii) It has high calorific values.

(iii) It is ~~pour~~ porous structure.

(iv)

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Q8 Describe Bomb Calorimeter?

Ans Bomb calorimeter is used for solid and non-volatile liquids for calculating the calorific values.

Principle: A known amount of fuel is burnt the heat produced is absorbed by known amount of water. The heat produced by unit mass/v is calculated.

Heat liberated = $m \times c = \Delta t (W + w)$

$$\boxed{c = \frac{\Delta t (W + w)}{m}}$$

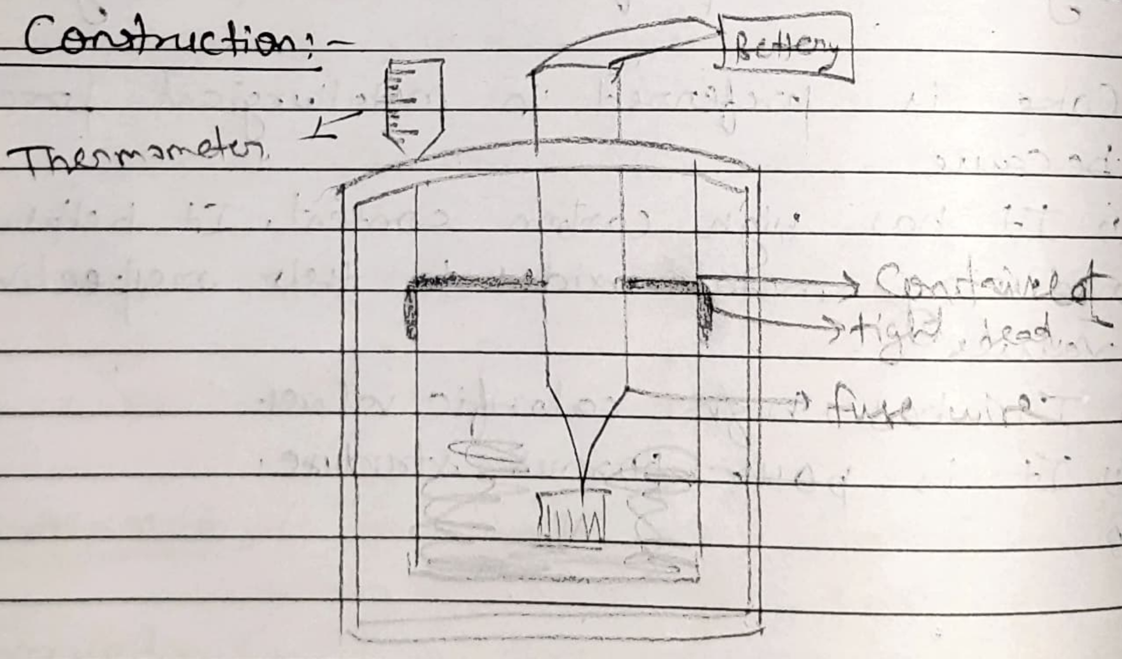
here Δt = diff in temp.

m = weight of fuel ~~ing~~ ignited

W = amount of water.

w = water eq. of calorimeter.

Construction:-



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Structure

(i) Stainless steel Bomb :- It contains given sample of fuel.

(ii) Copper calorimeter :- It contains a thermometer which is an electrically operated.

(iii) Air Jacket & water Jacket :- It cover & prevent any heat losses due to radiation.

Working :- A known mass of given fuel taken in crucible.

-> Crucible is supported over the ring.

-> lid is tightly screwed bomb is filled with O_2 at 25 atm. Fuze wire is stretched across the electrodes.

-> It's lowered down in copper calorimeter (water). It is connected to 6V battery & circuit is done. Now sample starts burning & heat is liberated & temp is measured.

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Q.9. What is synthetic Petrol? Explain Fischer-Tropsch method.

Ans. When coal & its product are hydrogenerated they converted into petroleum like substance at appropriate temp & pressure. This petrol like substance is known as synthetic petrol.

Fischer Tropsch process:-

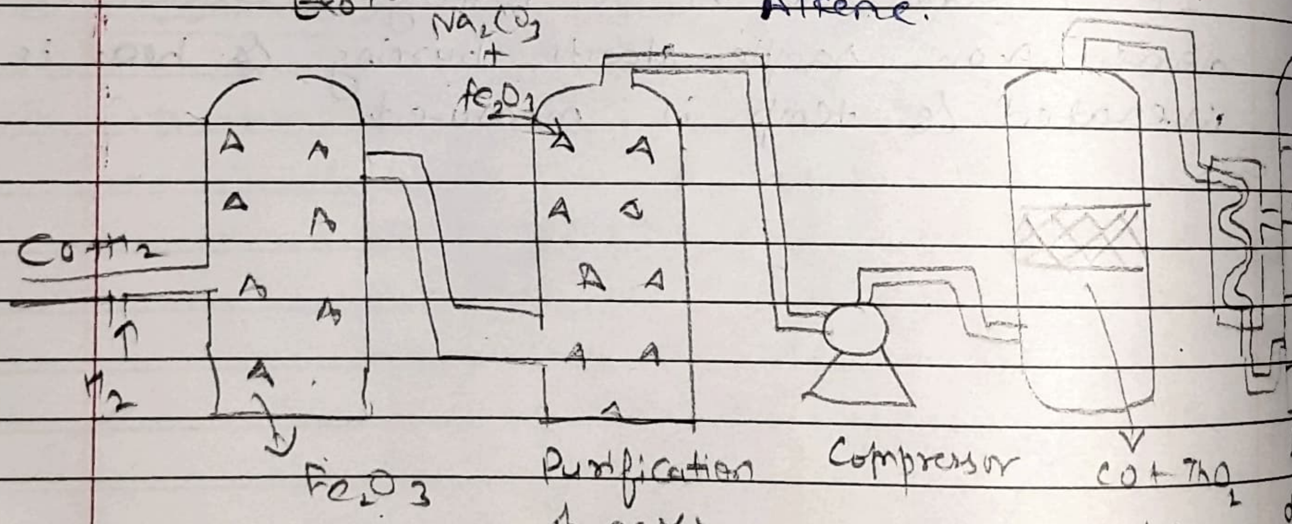
$CO + H_2$ (water gas) passed over catalyst at high temperature and pressure.
Zinc, Cr_2O_3, NiO_2 these are the catalyst which are used.

Reaction

$$nCO + 2nH_2 \longrightarrow \underset{\text{(Alkene)}}{C_nH_{2n}} + nH_2O + Heat$$

$$nCO + (2n+1)H_2 \longrightarrow \underset{\text{Alkene.}}{C_nH_{2n+2}} + nH_2O + Heat.$$

Exothermic



→ This is an expensive process.
→ Useful in emergency.

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Q10 What is reforming? Write its advantages?

Ans Reforming is used in petroleum & petrochemical industries to improve the quality of fuel by converting lower-grade hydrocarbons into more valuable products. It typically involves breaking & rearranging hydrocarbon molecules to increase the octane number of gasoline or produce hydrogen-rich gases.

Advantage

(i) It increases octane rating of fuel & reduces engine knocking.
(ii) Generates hydrogen, which is essential for refining processes like cracking, desulfurization.
(iii) Improve yield of valuable products.
(iv) Enhance fuel efficiency.
(v) Reduces environmental impact.

Q11 What is sweetening of Petrol?

Ans Sweetening is a refining process used to improve the quality of petrol (gasoline) by reducing its sulfur content & removing foul-smelling sulfur compound like mercaptans.

$$Na_2PbO_2 + 2RSH \longrightarrow Pb(SR)_2 + 2NaOH$$

$$Pb(SR)_2 + S \longrightarrow PbS + \underset{\text{Disulphide}}{RSSR}$$

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Q12. What is octane number. How it can be increased in fuel?

Ans Octane number is a measure of a fuel's ability to resist knocking or pre-detonation in an internal combustion engine. Higher octane number indicates better resistence of knocking & better quality of fuel.

Octane Number can be increased by several methods in fuel.

(i) Blending with high octane Components.
ex benzene, toluene, ethanol etc

(ii) Refining Process - like catalytic reforming, isomerization or alkylation.

(iii) Using octane Boosting Additives (Anti knocking agents) like TEL, MTBE, E10, E85 etc.

Q13 What is cracking, explain Fixed bed catalytic cracking?

Ans Decomposition of Higher hydrocarbon molecules into lighter and useful hydrocarbons

ex

$$\underset{\text{decane}}{C_{10}H_{22}} \xrightarrow[\text{cracking}]{\text{hydro}} \underset{\text{Pentane}}{C_5H_{12}} + \underset{\text{Pentene}}{C_5H_{10}}$$

It has two types —
(i) Thermal cracking (ii) Catalytic cracking

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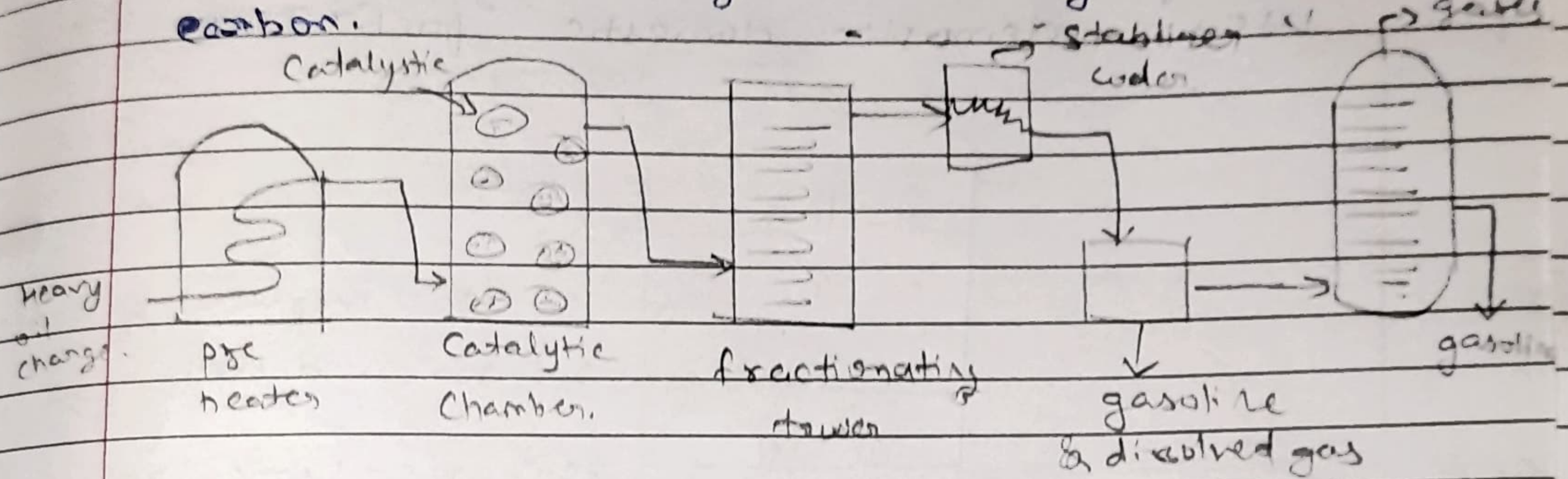
Fixed bed catalytic cracking :-

In this method Heavy oil vapours are preheated ~~catalyte~~ at 425° to 450°c and passed through catalytic chamber. About 40% of charged heavy oil is converted into gasoline along with 2-4% carbon.



Q14 Write a short note on fractional distillation?

Ans Crude oil is heated in a furnace at temp. of 400°c by which most of the oil is vaporised. These vapours are passed through distillation tower in which temp. decreased from ~~top~~ bottom to top ~~of~~ tower.

Principle The fraction having lowest boiling point condense near the top of the tower. Various fractions are collected according to their boiling points.

i) **Gasoline** → having low boiling point.
Calorific value = 11250 kcal/kg
Used as fuel in engines of automobile & aeroplanes.

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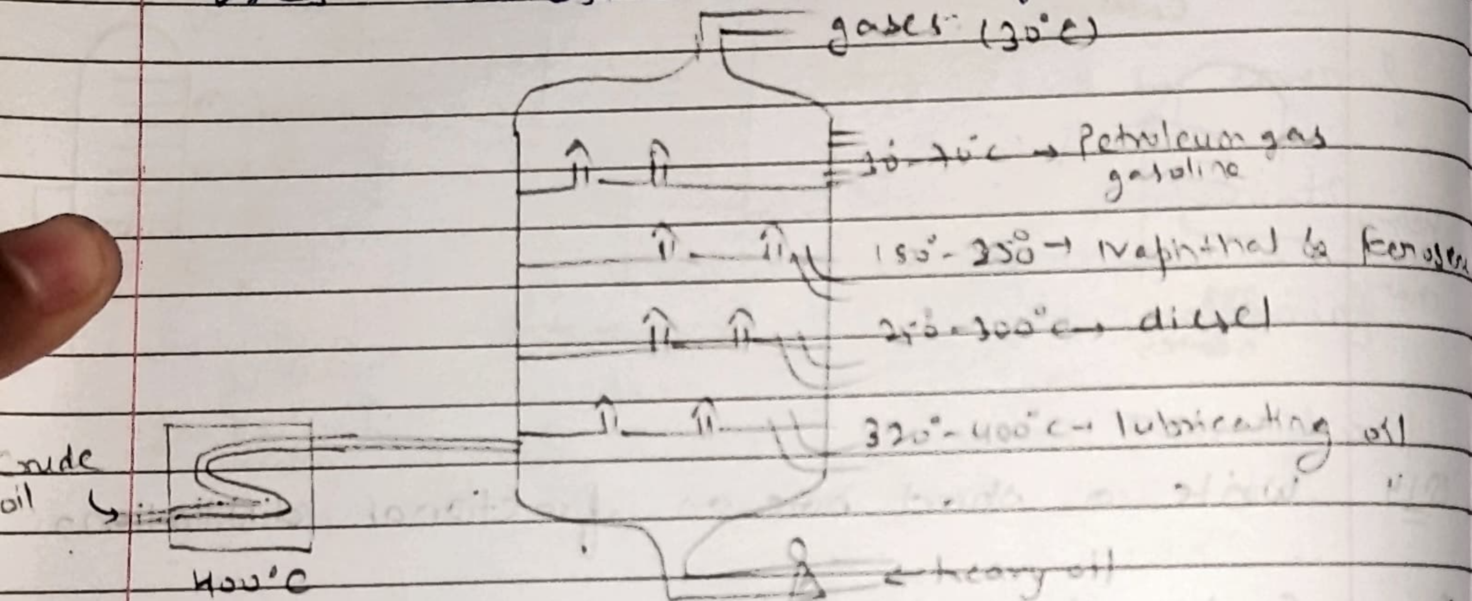
Kerosine oil calorific value = 11100 kcal/kg
used as good fuel for jet engines.

Liquified petroleum gases :- Light hydro-carbons such as propane & butane & can be stored in containers under pressure.
Calorific value = 27800 kcal/kg
uses - ~~demostic~~ domestic fuel.



Q15. What is producer gas? Explain Junker's calorimeter.